Useful Memory Has a Horizon

Temporal Validity, Roughness, and Structural Limits for Finite-Memory Tracking under Drift

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Abstract

Memory is not always an asset under drift. Past data reduce sampling noise, but as the target distribution moves, retaining more evidence also accumulates temporal misalignment with the present. Finite-memory distribution tracking is therefore governed by a temporal-validity problem: how long can retained evidence improve estimation of the present before drift-induced misalignment overtakes variance reduction?

A carrier-roughness useful-memory horizon law separates finite-memory error into a carrier term, coming from the chosen geometry and sample regime, and a roughness-controlled staleness term, coming from the drift path:

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le C_K n^{-a} + C_S \zeta n^H.$$

Here a is the carrier exponent and $H \in (0,1]$ is the temporal roughness exponent. Optimizing this balance yields the useful-memory scale $n^*(a,H) \asymp (C_K/\zeta)^{1/(a+H)}$, with the classical cube-root law appearing as the special case $a = 1/2$, $H = 1$.

We instantiate the law with a rigorous minimum kernel: a one-dimensional bounded-support fixed-span triangular array in which a quantile/Bahadur argument gives the canonical $a = 1/2$ carrier. A Gaussian witness lower bound shows that the resulting horizon is structural in this canonical regime. Beyond the minimum kernel, the same framework supports a Gaussian location minimax benchmark and a certified fixed-ε Sinkhorn operational frontier, showing that the same horizon logic remains meaningful outside the one-dimensional proof regime.

Under drift, memory is governed not by a single universal window law, but by the interaction between carrier geometry and temporal roughness. This viewpoint is adjacent to dynamic regret and adaptive windowing, but the object is different: temporal validity of retained evidence. Useful memory has a path-dependent horizon, and retained evidence can become invalid for the present before changepoint evidence becomes statistically visible.

Keywords: tracking under drift; useful memory; temporal validity; carrier-roughness useful-memory horizon law; invalidity gap; Wasserstein distance.
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1 Introduction

In stationary statistics, memory is an unambiguous resource: larger samples reduce variance because the target distribution does not change. Under drift, that logic fails. In streaming estimation, monitoring, and online adaptation, the evidence retained in memory was generated by distributions that may no longer represent the present. More memory therefore does two things at once: it reduces finite-sample noise, and it increases the age of the evidence used to estimate the current state.

That tension creates a horizon problem. Short windows are fresh but noisy. Long windows are stable but stale. The relevant question is not only whether memory helps, but how long retained evidence can remain temporally valid before drift-induced misalignment overtakes the remaining variance gain. The paper studies that question in distributional geometry and treats the horizon of useful memory as the main statistical object.

A carrier-roughness useful-memory horizon law captures this problem. The starting point is an abstract decomposition

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le C_K n^{-a} + C_S \zeta n^H,$$

where $a > 0$ is the finite-sample carrier exponent of the chosen geometry and sample regime, $H \in (0, 1]$ is the temporal roughness exponent of the drift path, and ζ is the roughness amplitude. Optimizing this balance yields the useful-memory scale

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

The cube-root law is recovered as the special case $a = 1/2$, $H = 1$, corresponding to a root-n carrier and worst-case linear staleness. The point of the paper is therefore not the cube-root law in isolation, but the larger object that explains when cube-root scaling appears, when it does not, and why the useful-memory horizon is path dependent.

To make this law concrete, the paper proceeds in three moves.

First, it identifies a *minimum kernel*: a one-dimensional bounded-support fixed-span triangular-array setting in which the carrier can be instantiated rigorously. In that regime, the quantile representation of W_2^2, a triangular-array Bahadur expansion, and a design-effect comparison yield the canonical carrier

$$\mathbb{E}\, W_2\Big(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\Big) = O(n^{-1/2}).$$

This supplies a concrete $a = 1/2$ regime of the abstract law and closes the canonical benchmark.

Second, it studies an *operational regime*: fixed-ε Sinkhorn geometry as an operational carrier whose finite-sample behavior remains useful even where raw Wasserstein geometry is dimensionally fragile. In the embedded fixed-span model studied here, this is more than a loose extension: exact support-complexity inheritance, the dual-complexity threshold $\alpha > k/2$, and calibrated ε-band stability together identify a certified operational region for the same horizon law.

Third, it formulates the broader noncanonical extension problem in the right way: not as a universal attempt to preserve the canonical root-n regime, but as a geometry-specific inheritance problem in which different regular families or operational geometries support different carrier exponents and therefore different members of the same horizon family.

The paper therefore has one central arc: close the canonical regime, identify a certified operational frontier beyond it, and state the broader family-specific extension target that replaces a false universal one. Memory under drift is governed by the interaction between carrier geometry and temporal roughness, not by a single universal window prescription.

The theory is complemented by a structural lower bound. On the canonical carrier regime $a = 1/2$, a Gaussian witness construction shows that the useful-memory scale is not merely the optimum of one procedure. The refined witness analysis also identifies the exact Gaussian testing frontier and shows that, for $H < 1$, the usual ramp witness is not the lowest-energy endpoint-saturating Hölder shape: the profile $g_r^{\min} = h^H - (h-r)^H$ minimizes witness energy on that class and strictly improves the ramp constant. The same lower-bound mechanism therefore supplies both the structural exponent law and a sharper picture of the canonical regime. In the Gaussian location benchmark, the same geometry can be pushed one step further to give a full minimax rate statement on the deterministic Hölder class itself. Beyond that canonical benchmark, the broader theorem target is family-specific rather than universal: identify which regular distribution families or operational geometries support which carrier exponent a, and hence which member of the horizon family they realize.

The empirical logic mirrors that arc. The paper first makes the basic variance–staleness trade-off visible through a finite-memory U-curve, then shows roughness-dependent horizon scaling, then maps the certified fixed-ε Sinkhorn frontier, and finally exhibits a measurable invalidity gap between temporal validity and changepoint evidence. The central conceptual distinction is simple: temporal validity is not changepoint evidence. A detector can remain statistically silent while retained evidence is already stale for the present.

This framing is adjacent to several literatures—dynamic regret, adaptive windowing, locally stationary smoothing, adaptive filtering, and distributional tracking under transport geometry—but the paper asks a different question. Dynamic-regret analysis asks how well one tracks a moving comparator, and adaptive-window or drift-detection methods ask when the accumulated evidence is sufficient to react. The question here is different: for finite-memory tracking under drift, how long does retained evidence remain temporally valid for estimating the present distribution before age becomes the dominant source of error?

The paper proceeds as follows. Section 2 develops the carrier-roughness useful-memory horizon law, closes the canonical regime, formulates the operational frontier, and states the broader regular-family extension target. Section 3 presents the corresponding finite-sample evidence. Section 4 discusses related work, and Section 5 discusses scope, limitations, and open problems.

2 The Carrier-Roughness Useful-Memory Horizon Law

Under drift, memory is both a statistical resource and a temporal liability. Retaining more data reduces finite-sample noise, but it also increases the age of the evidence used to represent the present. The object of interest is therefore not a detector threshold or a particular adaptive rule, but the error of finite-memory representation itself.

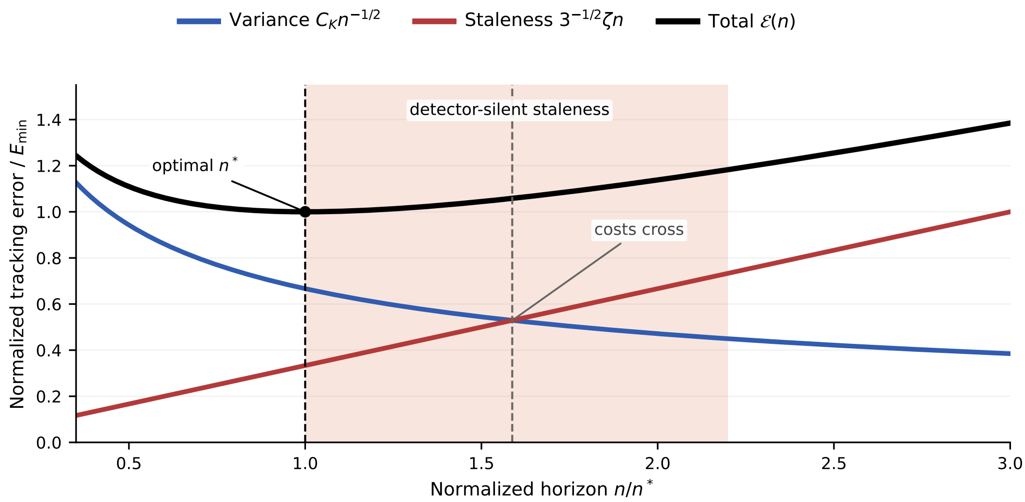

Figure 1: Two clocks of drift. The finite-sample term falls with horizon, the staleness term rises with age, and total error is minimized at the useful-memory horizon n^*. The shaded region represents detector-silent staleness: retained evidence can lose temporal validity before enough changepoint evidence accumulates to trigger an alarm.

For a finite-memory estimator with lag weights $w_0, \ldots, w_{n-1}$, the natural age variable is the effective lag

$$A(w) := \sum_{j=0}^{n-1} j\, w_j .$$

Its corresponding staleness at time t is the weighted transport misalignment

$$S_t(w) := \sum_{j=0}^{n-1} w_j\, W_2(P_{t-j}, P_t)\,.$$

For a uniform window, $A(w) = (n-1)/2$, so window length and temporal age are the same object up to constants. This is why memory length becomes a statistical question rather than a purely algorithmic one.

Throughout the paper, P_t denotes the present target law, $\bar{P}_t^{(n)}$ the window target obtained by averaging the last n target laws, and $\widehat{P}_t^{(n)}$ the estimator built from the last n observations. The carrier term always refers to the estimation error relative to $\bar{P}_t^{(n)}$, while the staleness term refers to the discrepancy between $\bar{P}_t^{(n)}$ and P_t.

The theory repeatedly combines three ingredients: a carrier benchmark relative to $\bar{P}_t^{(n)}$, a path-class control on the staleness between $\bar{P}_t^{(n)}$ and P_t, and, when available, a lower-bound or benchmark theorem showing that the resulting horizon is structural on the geometry under study.

2.1 Abstract Upper Law

The abstract theorem separates finite-sample carrier behavior from drift-induced staleness.

Theorem 2.1 (Abstract upper law)**.** *Let d be a probability metric satisfying the triangle inequality. At time t, let P_t be the present target law, let $\bar{P}_t^{(n)}$ be the window target, and let $\widehat{P}_t^{(n)}$ be the estimator built from the last n observations. Assume that for some constants $C_K > 0$, $C_S > 0$, exponent $a > 0$, roughness index $H \in (0, 1]$, and amplitude $\zeta > 0$,*

$$\mathbb{E}\, d\left(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\right) \le C_K n^{-a}$$

and

$$d\left(\bar{P}_t^{(n)}, P_t\right) \le C_S \zeta n^H$$

hold for all n in the regime of interest. Then

$$\mathbb{E}\, d\left(\widehat{P}_t^{(n)}, P_t\right) \le C_K n^{-a} + C_S \zeta n^H.$$

Proof. By the triangle inequality,

$$d\left(\widehat{P}_t^{(n)}, P_t\right) \le d\left(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\right) + d\left(\bar{P}_t^{(n)}, P_t\right).$$

Taking expectations and applying the two assumptions yields the claim. □

The proof is immediate, but the theorem makes the modular structure explicit: the path class supplies the staleness term, while the chosen geometry and sample regime supply the carrier term.

Corollary 2.2 (Optimized useful-memory horizon law)**.** *Under the assumptions of theorem 2.1, the upper envelope*

$$\Phi(n) := C_K n^{-a} + C_S \zeta n^H$$

is minimized, up to constant factors, at

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)},$$

and the corresponding optimized error scale is

$$\Phi(n^*) \asymp C_K^{H/(a+H)} \zeta^{a/(a+H)}.$$

Proof. Balance the decreasing carrier term against the increasing staleness term:

$$C_K n^{-a} \asymp C_S \zeta n^H.$$

This gives $n^{a+H} \asymp C_K/\zeta$, hence the displayed horizon law. Substituting that scale back into either term gives the optimized error scale. □

The cube-root law is recovered when $a = 1/2$ and $H = 1$. More generally, the carrier exponent and the roughness exponent jointly determine the useful-memory scale. There is no universal horizon independent of those two ingredients.

2.2 Path Classes and the Roughness Index

The staleness side of the law is controlled by a deterministic path class. For $H \in (0, 1]$ and $\zeta > 0$, write

$$\mathfrak{P}_H(\zeta) := \left\{ (P_s)_{s\in\mathbb{Z}} : W_2(P_{t-j}, P_t) \le \zeta j^H \text{ for all } j \ge 0 \right\}.$$

The case $H = 1$ is the local Lipschitz envelope; smaller H correspond to slower accumulation of staleness with lag.

Lemma 2.3 (Hölder staleness bound via averaged couplings)**.** *Let*

$$\bar{P}_t^{(n)} := \frac{1}{n}\sum_{j=0}^{n-1} P_{t-j}.$$

If the target path belongs to $\mathfrak{P}_H(\zeta)$*, then*

$$W_2^2\left(\bar{P}_t^{(n)}, P_t\right) \le \frac{\zeta^2}{n}\sum_{j=0}^{n-1} j^{2H},$$

and therefore

$$W_2\left(\bar{P}_t^{(n)}, P_t\right) \le \zeta \left(\frac{1}{n}\sum_{j=0}^{n-1} j^{2H}\right)^{1/2} = C_{H,n}\,\zeta n^H,$$

where

$$C_{H,n} := \left(n^{-(2H+1)}\sum_{j=0}^{n-1} j^{2H}\right)^{1/2}.$$

In particular, $C_{H,n} \to (2H+1)^{-1/2}$ *as* $n \to \infty$*.*

Proof. For each lag j, let π_j be an optimal coupling between P_{t-j} and P_t. Their average

$$\bar{\pi} := \frac{1}{n}\sum_{j=0}^{n-1} \pi_j$$

couples $\bar{P}_t^{(n)}$ with P_t, so

$$W_2^2\left(\bar{P}_t^{(n)}, P_t\right) \le \int \|x-y\|^2\, d\bar{\pi}(x,y) = \frac{1}{n}\sum_{j=0}^{n-1} W_2^2(P_{t-j}, P_t).$$

Applying the Hölder envelope gives the displayed finite-n constant, and taking square roots completes the proof. □

Combining lemma 2.3 with corollary 2.2 yields the family

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

On the canonical statistical regime $a = 1/2$,

$$n^*(1/2, H) \asymp (C_K/\zeta)^{2/(1+2H)}.$$

At $H = 1$, this recovers the cube-root horizon; at $H = 1/2$, it yields the intermediate $n^* \asymp C_K/\zeta$ regime. These are members of the same law, not separate theorem stories. The finite-n constant in lemma 2.3 is the constant used throughout the paper for uniform windows; the simpler $(2H+1)^{-1/2}$ expression is its large-window limit.

2.3 Minimum Kernel: A Rigorous Carrier Instantiation

The abstract law needs at least one rigorous carrier instantiation. The minimum kernel is the bounded-support fixed-span one-dimensional triangular array. It provides the first rigorous carrier entering the abstract horizon law.

Proposition 2.4 (Minimum-kernel carrier)**.** *Let $X_{n,1}, \dots, X_{n,n}$ be independent one-dimensional observations with laws $P_{n,1}, \dots, P_{n,n}$, and define the window mixture*

$$\bar{P}_n = \frac{1}{n} \sum_{j=1}^{n} P_{n,j}.$$

Assume:

(i) all $P_{n,j}$ are supported on a common compact interval $[L, U]$;

(ii) the within-window drift span is uniformly bounded in n;

(iii) on an interior quantile band $u \in [\varepsilon, 1-\varepsilon]$, the mixture density $\bar{f}_n$ is bounded below and uniformly Hölder;

(iv) the triangular-array empirical quantile process admits a Bahadur representation with integrated remainder $o(n^{-1})$ on that interior band.

Then

$$\mathbb{E}\, W_2^2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) = O(n^{-1}),$$

and therefore

$$\mathbb{E}\, W_2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) = O(n^{-1/2}).$$

Proof sketch. In one dimension,

$$W_2^2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) = \int_0^1 \left(\widehat{q}_n(u) - q_n(u)\right)^2 du,$$

so the problem reduces to the integrated L_2 error of the empirical quantile process. On the interior band, the Bahadur representation gives

$$\widehat{q}_n(u) - q_n(u) = \frac{u - \widehat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u).$$

The leading term has integrated variance $O(n^{-1})$ because the observations are independent and the fixed-span triangular design changes only constants relative to the i.i.d. mixture benchmark. The remainder splits into an empirical increment term and a Taylor term; the empirical increment term is the dominant one, but under the stated interior regularity it still satisfies an integrated $o(n^{-1})$ bound. Bounded support makes the boundary-band contribution lower-order. Summing the leading term, the lower-order cross term, and the integrated remainder gives $\mathbb{E}W_2^2 = O(n^{-1})$, and Jensen yields the displayed W_2 rate.

The main conclusion is the exponent. Under this bounded-support fixed-span regime, the triangular-array carrier remains root-n and differs from the i.i.d. mixture benchmark only through a design effect.

Remark 2.5 (A self-contained sufficient package). The nonstandard assumption in proposition 2.4 is item (iv). A practical sufficient package for the regime intended here is the following: independence across the row, common bounded support, uniformly bounded within-window span, mixture quantiles that stay on an interior band where $\bar{f}_n$ is bounded below and uniformly Hölder, and row-wise perturbations of the i.i.d. mixture benchmark that change the empirical-process variance only by a fixed-span design effect. In that setting, the usual one-dimensional Bahadur expansion for smooth quantile functionals yields an integrated remainder $o(n^{-1})$. This covers, for example, bounded-support smooth location perturbations of a common base density and bounded smooth monotone transforms of such a base law, provided the fixed-span condition is maintained.

2.4 A Structural Lower Bound

The upper law would be substantially weaker if it were only the optimum of one estimator. The lower bound shows that, on the canonical carrier regime $a = 1/2$, the useful-memory scale is imposed by the drift geometry itself.

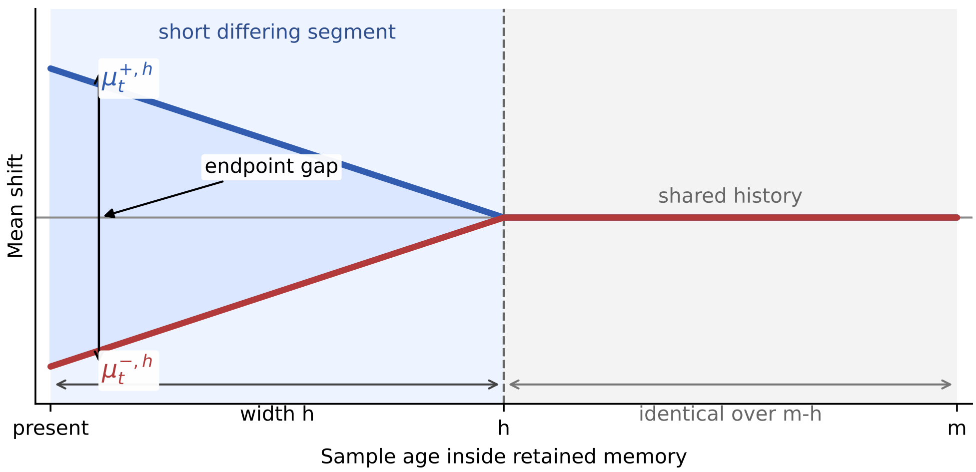

Figure 2: Gaussian witness for the structural lower bound. The two paths share most of the retained history and differ only on a recent segment of width h, yet remain separated at the present. Optimizing that recent segment already produces the same horizon exponent as the upper law on the canonical carrier regime.

Proposition 2.6 (Structural lower bound on the canonical carrier regime)**.** *Fix $H \in (0, 1]$ and work on the Gaussian location subclass with known variance σ^2 and witness paths*

$$\mu_{t-j}^{\pm,h} = \pm\beta(h - j)_+^H, \qquad j = 0, 1, \ldots, h,$$

where $\beta \leq \zeta$ enforces the roughness budget. Let $\mathcal{G}_{H,\zeta}^{\mathrm{Gauss}}$ denote the corresponding Gaussian location subclass of target paths. Then there exists a constant $c_H > 0$ depending only on H such that the endpoint tracking risk over this subclass satisfies

$$\inf_{\widehat{P}} \sup_{P \in \mathcal{G}_{H,\zeta}^{\mathrm{Gauss}}} \mathbb{E}\, W_2\left(\widehat{P}_t, P_t\right) \geq c_H\, \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Consequently, on the canonical carrier regime $a = 1/2$, the useful-memory scale

$$h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$$

is structural even on this restricted subclass.

Proof sketch. Compare two hypotheses generated by the witness paths $\mu_{t-j}^{+,h}$ and $\mu_{t-j}^{-,h}$. At the present time, the endpoint separation is of order βh^H. On the Gaussian location subclass, this is also the W_2 separation. The Kullback–Leibler divergence between the two hypotheses is proportional to

$$\beta^2 \sum_{r=1}^{h} r^{2H}/\sigma^2.$$

Choosing

$$\beta = \min\left\{ \zeta, \frac{\sigma}{2\sqrt{\sum_{r=1}^{h} r^{2H}}} \right\}$$

keeps the testing problem in the Le Cam regime while respecting the roughness budget. The resulting two-point lower bound is of order βh^H. Balancing the active roughness constraint against the testing constraint gives $h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$ and therefore the displayed lower law.

At $H = 1$, this recovers the cube-root horizon and the lower scale $\sigma^{2/3}\zeta^{1/3}$. The lower bound is subclass based rather than class-tight, but that is enough for the central claim: the horizon is not just an estimator artifact.

The proposition above is the lower-bound fact needed for the main horizon law. The same witness analysis also exposes a sharper constant-level geometry. We record those refinements next because they explain what is intrinsic to the canonical $a = 1/2$ regime and what was only an artifact of the original ramp proof with its testing relaxation.

2.5 Refined Geometry of the Gaussian Witness

Let Φ and φ denote the standard normal distribution function and density. For the ramp witness in proposition 2.6, exact Gaussian testing gives a one-dimensional variational problem rather than a Pinsker-style constant.

Proposition 2.7 (Exact Gaussian ramp frontier)**.** *Fix $H \in (0, 1]$ and set*

$$p_H := \frac{2H}{2H+1}.$$

Let $x_H > 0$ be the unique maximizer of $x^{p_H}\Phi(-x)$, equivalently the solution of

$$p_H \Phi(-x_H) = x_H \varphi(x_H).$$

For the Gaussian ramp witness

$$\mu_{t-j}^{\pm,h} = \pm\beta(h-j)_+^H,$$

the exact two-point lower bound has asymptotic constant

$$C_H^{\mathrm{ramp}} = (2H+1)^{H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H),$$

with horizon shape

$$h_H^* \sim (\sqrt{2H+1}\, x_H)^{2/(2H+1)} \left(\frac{\sigma}{\zeta}\right)^{2/(2H+1)}.$$

Proof sketch. For fixed h, the likelihood-ratio testing problem between the two Gaussian paths depends on

$$S_h = \sum_{r=1}^{h} r^{2H}$$

and has exact Bayes error $\Phi(-\beta\sqrt{S_h}/\sigma)$. The endpoint loss is of order βh^H, so the witness objective is

$$L_h(\beta) = \beta h^H \Phi(-\beta\sqrt{S_h}/\sigma).$$

In the large-ratio regime $\sigma/\zeta \to \infty$, use $S_h \sim h^{2H+1}/(2H+1)$ and set $x = \zeta\sqrt{S_h}/\sigma$. After normalizing by $\sigma^{2H/(2H+1)}\zeta^{1/(2H+1)}$, the objective becomes $(2H+1)^{H/(2H+1)} x^{p_H} \Phi(-x)$. Differentiating gives the stated scalar equation and constant. □

The ramp is therefore a valid structural witness, but it is not the best endpoint-saturating Hölder shape when $H < 1$.

Proposition 2.8 (Endpoint-minimal witness as the energy minimizer)**.** *Fix $H \in (0,1]$ and $h \geq 1$. Among all discrete profiles $g = (g_0, \ldots, g_h)$ satisfying*

$$g_0 = 0, \qquad g_h = h^H, \qquad |g_r - g_s| \leq |r-s|^H,$$

the profile

$$g_r^{\min} = h^H - (h-r)^H$$

minimizes the testing energy $\sum_{r=1}^{h} g_r^2$. Its asymptotic energy constant is

$$I_H = \int_0^1 \left(1-(1-u)^H\right)^2 du = \frac{2H^2}{(H+1)(2H+1)}.$$

Consequently the exact Gaussian endpoint-minimal witness has constant

$$C_H^{\min} = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H),$$

where x_H is the root from proposition 2.7. For $H < 1$, this strictly improves the ramp constant by the factor

$$\left(\frac{H+1}{2H^2}\right)^{H/(2H+1)}.$$

Proof. For any feasible profile, the endpoint constraint and Hölder condition imply

$$g_r \geq h^H - (h-r)^H.$$

The displayed $g^{\min}$ attains this lower envelope and is feasible because $x \mapsto x^H$ is subadditive on $(0, \infty)$ for $H \leq 1$. It is therefore pointwise dominated by every feasible profile and minimizes $\sum_{r=1}^{h} g_r^2$. The Riemann-sum limit gives

$$h^{-(2H+1)} \sum_{r=1}^{h} \left(h^H - (h-r)^H\right)^2 \to I_H.$$

Substituting a general energy constant I into the exact Gaussian reduction gives $I^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H)$. Taking $I = I_H$ yields the endpoint-minimal constant; comparing it with the ramp energy constant $(2H+1)^{-1}$ gives the improvement factor. At $H = 1$, the two profiles coincide. □

These refinements matter for more than witness design. They also change the constant-level interpretation of the canonical lower regime. Once the roughness side is expressed through the exact uniform-window staleness constant and the lower side uses the exact Gaussian witness constants, the remaining proof gap has a clean structural form.

Proposition 2.9 (Refined proof-gap consequence on the canonical witness regime)**.** *Fix $H \in (0, 1]$ and let*

$$C_S^{\mathrm{asymp}}(H) := (2H+1)^{-1/2}.$$

For $a > 0$, define

$$\Sigma(a, H) := \left(\frac{a}{a+H}\right)^{-a/(a+H)} + \left(\frac{H}{a+H}\right)^{-H/(a+H)}.$$

For either refined witness constant

$$C_H^{\mathrm{ref}} \in \{C_H^{\mathrm{ramp}}, C_H^{\min}\},$$

consider the associated constant-level proof-gap formula

$$\Gamma_{\mathrm{ref}}(a, H) = \Sigma(a, H) \left(\frac{C_S^{\mathrm{asymp}}(H)}{C_H^{\mathrm{ref}}}\right)^{a/(a+H)}.$$

Then, for every fixed $H \in (0, 1]$,

$$\inf_{a>0} \Gamma_{\mathrm{ramp}}(a, H) = \inf_{a>0} \Gamma_{\min}(a, H) = 2,$$

and the infimum is attained only in the boundary limit $a \downarrow 0$.

Proof sketch. Write

$$p = \frac{2H}{2H+1} \in (0, 2/3].$$

For the ramp witness, the scalar root equation

$$p\Phi(-x_H) = x_H \varphi(x_H)$$

together with the Komatsu-style bound

$$\Phi(-x) \leq \frac{2\varphi(x)}{x + \sqrt{x^2 + 8/\pi}}$$

gives the explicit lower envelope

$$\frac{C_S^{\mathrm{asymp}}(H)}{C_H^{\mathrm{ramp}}} \geq \sqrt{2\pi}\frac{(1-p)^{(p+1)/2}(p+2/\pi)^{(p+1)/2}}{p^p}.$$

Splitting at $p_0 = 1 - 2/\pi$ (equivalently $H_0 = \pi/4 - 1/2$) reduces this lower envelope to two elementary log-concave functions whose endpoint values are all strictly larger than 1. Hence

$$C_S^{\mathrm{asymp}}(H) > C_H^{\mathrm{ramp}} \qquad \text{for all } H \in (0,1].$$

For the endpoint-minimal witness, the same tail bound gives

$$\frac{C_S^{\mathrm{asymp}}(H)}{C_H^{\min}} \geq \sqrt{2\pi}\,\sqrt{1-p}\frac{(p+2/\pi)^{(p+1)/2}}{(2-p)^{p/2}}.$$

The logarithmic derivative of this lower envelope admits an explicit upper bound whose numerator is a cubic polynomial in p with negative derivative on $p \in (0, 2/3]$. The lower envelope is therefore strictly decreasing, so its minimum is forced to the endpoint $p = 2/3$, where it is still strictly larger than 1. Hence

$$C_S^{\mathrm{asymp}}(H) > C_H^{\min} \qquad \text{for all } H \in (0,1].$$

For either refined constant, set

$$R_{\mathrm{ref}}(H) = C_S^{\mathrm{asymp}}(H)/C_H^{\mathrm{ref}} > 1.$$

Then

$$\Gamma_{\mathrm{ref}}(a,H) = \Sigma(a,H)R_{\mathrm{ref}}(H)^{a/(a+H)}.$$

Since $\Sigma(a,H) \geq 2$ and $R_{\mathrm{ref}}(H)^{a/(a+H)} \geq 1$, one has $\Gamma_{\mathrm{ref}}(a,H) \geq 2$. As $a \downarrow 0$, the exponent $a/(a+H) \downarrow 0$ and $\Sigma(a,H) \to 2$, so $\Gamma_{\mathrm{ref}}(a,H) \to 2$. This yields the displayed infimum formula. □

This proposition has the same scope as the refined witness geometry itself. It does not claim a class-tight minimax theorem for all deterministic Hölder paths. Its role is narrower and cleaner: within the exact Gaussian witness program, the conservative constant from the older testing relaxation is replaced by witness-level constants for which the residual gap is exactly the structural factor-two crossover between additive upper and two-point lower mechanisms.

The same refined witness geometry also yields a full class-level benchmark once the model is specialized to Gaussian location tracking. In that setting, the deterministic Hölder envelope, the finite-memory estimator, and the lower-bound witness all live in the same one-dimensional parameterization, so the canonical regime admits a direct minimax rate statement.

Theorem 2.10 (Gaussian location minimax benchmark on the deterministic Hölder class)**.** *Fix $H \in (0,1]$, $\zeta > 0$, and $\sigma > 0$. Let $\mathfrak{M}_H(\zeta)$ denote the class of deterministic mean paths $(\mu_s)_{s\in\mathbb{Z}}$ such that*

$$|\mu_{t-j} - \mu_t| \leq \zeta j^H \qquad \textit{for all } j \geq 0.$$

Suppose the observations are independent with

$$Y_{t-j} \sim N(\mu_{t-j}, \sigma^2), \qquad j = 0, 1, \ldots, n-1,$$

and let the target law at time t be $P_t = N(\mu_t, \sigma^2)$. Define the optimized finite-memory minimax risk

$$\mathcal{R}_H^*(\sigma,\zeta) \coloneqq \inf_{n\geq 1} \inf_{\widehat{P}_t^{(n)}} \sup_{\mu\in\mathfrak{M}_H(\zeta)} \mathbb{E}\, W_2\Big(\widehat{P}_t^{(n)}, P_t\Big).$$

Then there exist constants $0 < c_H \leq C_H < \infty$, depending only on H, such that

$$c_H\, \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)} \leq \mathcal{R}_H^*(\sigma,\zeta) \leq C_H\, \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

In particular, the deterministic Hölder class has the canonical minimax horizon scale

$$n_H^* \asymp (\sigma/\zeta)^{2/(2H+1)}.$$

Moreover, the uniform-window mean estimator has an exact asymptotic upper constant on this benchmark. If $y_H > 0$ solves

$$H y_H \big(2\Phi(y_H) - 1\big) = \varphi(y_H),$$

then its asymptotic horizon shape is

$$A_H^{\mathrm{mean}} = ((H+1)y_H)^{2/(2H+1)}$$

and its asymptotic risk constant is

$$U_H^{\mathrm{mean}} = ((H+1)y_H)^{-1/(2H+1)} \Big(2\varphi(y_H) + y_H\big(2\Phi(y_H) - 1\big)\Big).$$

Proof sketch. For the upper bound, estimate the present mean by the uniform-window average

$$\widehat{\mu}_t^{(n)} \coloneqq \frac{1}{n}\sum_{j=0}^{n-1} Y_{t-j}$$

and set $\widehat{P}_t^{(n)} = N(\widehat{\mu}_t^{(n)}, \sigma^2)$. Because equal-variance Gaussian laws satisfy

$$W_2\big(N(m_1,\sigma^2), N(m_2,\sigma^2)\big) = |m_1 - m_2|,$$

the risk reduces to mean estimation error. The noise term is $\sigma\sqrt{2/(\pi n)}$, while the deterministic bias is bounded by

$$\frac{\zeta}{n}\sum_{j=0}^{n-1} j^H,$$

so

$$\sup_{\mu\in\mathfrak{M}_H(\zeta)} \mathbb{E}\, W_2\Big(\widehat{P}_t^{(n)}, P_t\Big) \leq \sigma\sqrt{\frac{2}{\pi n}} + \frac{\zeta}{n}\sum_{j=0}^{n-1} j^H.$$

Rescaling by $n = A(\sigma/\zeta)^{2/(2H+1)}$ yields a one-parameter normalized upper envelope. Differentiating its exact Gaussian-location form gives the stated root equation for y_H, and therefore the displayed asymptotic upper constant and horizon shape. In particular, this proves the stated upper rate and horizon scale.

For the lower bound, use the symmetric two-point subfamily

$$\mu_{t-j}^{\pm,h} = \pm\beta\big(h^H - j^H\big)_+, \qquad j \geq 0,$$

with $\beta \le \zeta$. This family lies in $\mathfrak{M}_H(\zeta)$ because

$$|\mu_t^{\pm,h} - \mu_{t-j}^{\pm,h}| = \beta \min(h^H, j^H) \le \zeta j^H.$$

Moreover, for fixed endpoint value βh^H, it minimizes the Gaussian testing energy over all symmetric lag profiles allowed by the present-point Hölder envelope, since each lag coordinate is independently minimized by $\beta(h^H - j^H)_+$. The exact Gaussian two-point calculation from the refined witness analysis therefore applies on the full deterministic Hölder class and yields a lower bound of order

$$\sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Optimizing over h gives the same horizon scale $(\sigma/\zeta)^{2/(2H+1)}$. Combining the two sides proves the theorem. □

2.6 Beyond the Minimum Kernel

The one-dimensional minimum kernel is the first rigorous carrier instantiation. Two broader settings are also important for interpreting the law.

Extended regime. In low-dimensional or low-intrinsic-dimensional settings, the natural question is whether the triangular-array carrier inherits the same exponent as the i.i.d. mixture benchmark up to a constant-level gap. The clearest evidence comes from embedded $k = 1$ support inside a larger ambient space, where triangular and i.i.d. slopes remain near the $a = 1/2$ regime under fixed span.

Operational regime. For a practical geometry such as fixed-ε debiased Sinkhorn, the corresponding question is whether the induced finite-sample carrier is stable enough to enter the same horizon law. The experiments reported below support that interpretation: on embedded low-intrinsic-dimensional support, the triangular and i.i.d. benchmark carriers remain close across the tested regularization values, with ε affecting constants more than the qualitative carrier exponent. The relevant theorem in this regime therefore does not take the form of a raw-W_2 statement in arbitrary dimension, but of an operational carrier theorem of the form

$$\mathbb{E}\, S_\varepsilon\left(\widehat{P}_{t,\mathrm{tri}}^{(n)}, \bar{P}_t^{(n)}\right) \le C_\varepsilon n^{-a_\varepsilon}$$

under bounded support, fixed span, and structural low-dimensionality, where S_ε denotes a fixed-ε Sinkhorn-type geometry and the exponent a_ε matches the corresponding i.i.d. mixture benchmark. For fixed-ε entropic transport quantities, theorem-level i.i.d. results already point in this direction. Stromme proves minimum-intrinsic- dimension scaling for Sinkhorn divergence values under bounded Lipschitz costs, with the statistical error governed by the smaller covering complexity of the two measures [Stromme, 2023, Theorem 3.1]. Groppe and Hundrieser prove lower- complexity adaptation for empirical entropic OT costs and show that the finite- iteration Sinkhorn algorithm preserves the same statistical order up to an explicit computational tolerance [Groppe and Hundrieser, 2024, Theorem 4.10 and Remark 4.11]. On the embedded fixed-span model studied here, the remaining structural pieces can also be isolated cleanly: the triangular-window and i.i.d.-mixture supports have identical axis-aligned covering complexity at operational scale, and with squared-Euclidean cost the dual smoothness condition reduces to the LCA threshold $\alpha > k/2$. This makes the operational question structurally sharp rather than amorphous: once the triangular-vs.-i.i.d. carrier gap is calibrated on a regularization band, the horizon law can be stated on that certified region.

Definition 2.11 (Certified operational region)**.** Fix a bounded embedded fixed-span model, an intrinsic complexity index k, a dual Hölder smoothness level α, and an operational band $\mathcal{B} \subset (0, \infty)$ of regularization values. We call $(k, \mathcal{B})$ a certified operational region for fixed-ε Sinkhorn if:

(i) the i.i.d. fixed-ε benchmark on the same support class admits a carrier bound with exponent a_ε on $\mathcal{B}$;

(ii) the triangular-window and i.i.d.-mixture supports have identical operational covering complexity on $\mathcal{B}$;

(iii) the dual class lies in the parametric adaptation region $\alpha > k/2$;

(iv) across $\mathcal{B}$, the calibrated triangular and i.i.d. carrier slopes stay useful and uniformly close.

Proposition 2.12 (Certified operational frontier)**.** *On the bounded embedded fixed-span model with squared-Euclidean cost, the first three items in definition 2.11 are structural: the support-complexity inheritance is exact, and the dual-complexity condition is precisely the LCA threshold $\alpha > k/2$. Consequently, any calibrated band $\mathcal{B}$ satisfying item (iv) supports the carrier-roughness horizon law on a certified operational region. In particular, on bands where the i.i.d. benchmark stays in the parametric Sinkhorn regime, the operational carrier takes the canonical form*

$$\mathbb{E}\, S_\varepsilon\left(\widehat{P}^{(n)}_{t,\mathrm{tri}}, \bar{P}^{(n)}_t\right) \le C_\varepsilon n^{-1/2}, \qquad \varepsilon \in \mathcal{B},$$

and the corresponding useful-memory scale is

$$n^*_\varepsilon(H) \asymp (C_\varepsilon/\zeta)^{1/(H+1/2)}.$$

More generally, if the i.i.d. benchmark on $\mathcal{B}$ has exponent a_ε, then the operational horizon law on that region is

$$n^*_\varepsilon(a_\varepsilon, H) \asymp (C_\varepsilon/\zeta)^{1/(a_\varepsilon+H)}.$$

Proof sketch. The i.i.d. side is the benchmark supplied by Stromme [2023] and Groppe and Hundrieser [2024]. In the present embedded fixed-span model, the triangular-window and i.i.d.-mixture supports have identical operational covering bounds, so the support-complexity side of the inheritance problem is already closed. The squared-Euclidean chart model is smooth enough that the dual-complexity condition reduces to $\alpha > k/2$, which is exactly the parametric LCA region. Once the triangular-vs.-i.i.d. carrier gap is calibrated to remain uniformly small on $\mathcal{B}$, the operational carrier enters the abstract upper law just like any other carrier exponent. Optimizing the resulting upper envelope gives the displayed horizon scale.

This proposition isolates the structural part of the operational story and the point at which calibration still enters. It does not yet prove a full triangular-array Sinkhorn theorem with explicit uniform constants across $\mathcal{B}$; rather, it identifies a certified region in which the abstract horizon law is justified by a closed i.i.d. benchmark, an exact complexity inheritance statement, and a calibrated triangular-vs.-benchmark comparison.

Corollary 2.13 (Certified fixed-ε operational region)**.** *In the synthetic embedded fixed-span calibration used in this paper, with $\alpha = 2$ and span 0.25, the maximal certified regularization bands are:*

$$\varepsilon_{\max} = 0.5 \text{ for } (d,k) \in \{(8,1),(8,2),(12,2)\}, \qquad \varepsilon_{\max} = 0.2 \text{ for } (d,k) = (12,1).$$

Thus the operational frontier is not a dimension-free blanket statement. It is a certified region inside the fixed-span model on which the practical Sinkhorn carrier inherits the useful-memory law of the corresponding i.i.d. benchmark.

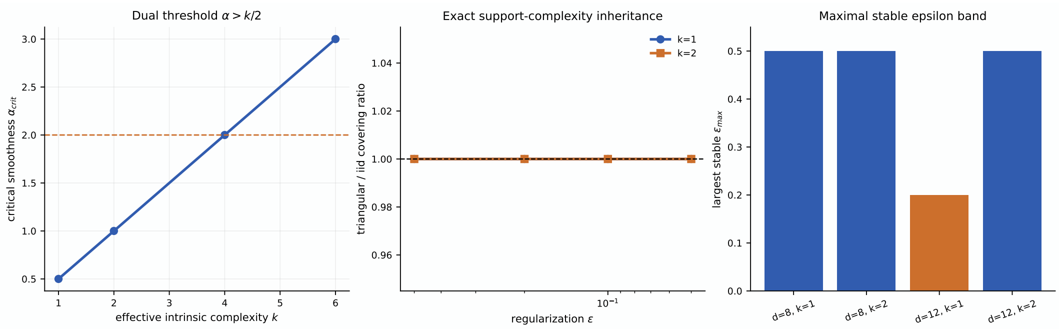

Figure 3: Certified operational frontier. Left: the dual threshold $\alpha > k/2$ that identifies the parametric adaptation regime. Middle: exact support-complexity inheritance in the embedded fixed-span model. Right: maximal certified ε-bands in the calibrated synthetic setting. Together these three ingredients isolate the operational region on which fixed-ε Sinkhorn supports the same carrier-roughness horizon law as its i.i.d. benchmark.

Regular-family extension target. The noncanonical question is broader than preserving the root-n carrier. The relevant objective is to identify distribution families and geometries that support some member of the carrier-roughness horizon family, with carrier exponent $a = a(\mathcal{G})$ determined by the geometry rather than fixed a priori. A natural candidate class is a locally regular dominated family: a parameterized family $\{P_\theta : \theta \in \Theta\}$ with a common dominating measure, locally quadratic testing geometry,

$$D_{\mathrm{KL}}(P_\theta, P_{\theta'}) \asymp \|\theta - \theta'\|^2,$$

and local comparability between parameter displacement and the chosen metric,

$$d(P_\theta, P_{\theta'}) \asymp \|\theta - \theta'\|^\alpha$$

for some metric exponent $\alpha > 0$. In such a family, a deterministic Hölder envelope in parameter space transfers to a deterministic staleness law in the observation geometry with metric roughness exponent αH, while the i.i.d. carrier problem remains regular enough for Le Cam-style lower arguments and benchmark upper laws to interact on a common local scale.

Conjecture 2.14 (Regular-family horizon inheritance)**.** *Let d be a probability geometry and let $\{P_\theta : \theta \in \Theta\}$ be a locally regular dominated family for which the i.i.d. benchmark satisfies a metric carrier law*

$$\mathbb{E}\, d\Big(\widehat{P}^{(n)}_{\mathrm{iid}}, P_\theta\Big) \le C_K n^{-a}$$

uniformly on the parameter region of interest. Suppose the path obeys a deterministic Hölder envelope in parameter space,

$$\|\theta_{t-j} - \theta_t\| \le \zeta j^H.$$

Then the induced metric staleness should scale as

$$d(P_{\theta_{t-j}}, P_{\theta_t}) \lesssim \zeta^\alpha j^{\alpha H},$$

so the corresponding tracking problem should satisfy the horizon-family member

$$n^*(a, \alpha H) \asymp (C_K/\zeta^\alpha)^{1/(a+\alpha H)},$$

with optimized risk of order

$$C_K^{\alpha H/(a+\alpha H)} \zeta^{\alpha a/(a+\alpha H)}.$$

In the locally regular parametric subcase, the natural metric carrier exponent is $a = \alpha/2$, giving the horizon scale

$$n^*(\alpha/2, \alpha H) \asymp (C_K/\zeta^\alpha)^{1/(\alpha(1/2+H))}.$$

This figure makes the noncanonical point concrete. The right extension problem is not to preserve the canonical horizon law for every family, but to identify which family geometry induces which carrier exponent and therefore which member of the horizon family.

The distinction between regular, support-changing, and singular families shows why such a regularity package is necessary. Continuous one-dimensional scale families such as Gaussian scale and uniform scale are broadly compatible with the same horizon exponent, but their local testing geometry differs sharply: Gaussian scale is regular, whereas uniform scale has support-driven nonregularity and atomic scale families become singular. So a full distributional extension should be formulated for regular dominated families or operational geometries with comparable local behavior, not for an arbitrary bare W_2-Hölder path class.

Remark 2.15 (Local-geometry taxonomy)*.* Gaussian-like regular families have locally quadratic testing geometry and are the clearest theorem targets for distributional horizon inheritance. Support-changing families can preserve the same transport scaling while losing local quadratic regularity, so they require a different lower theory. Atomic or singular families are genuine obstruction classes: they show that transport-Hölder control alone is too weak to support a single universal extension theorem.

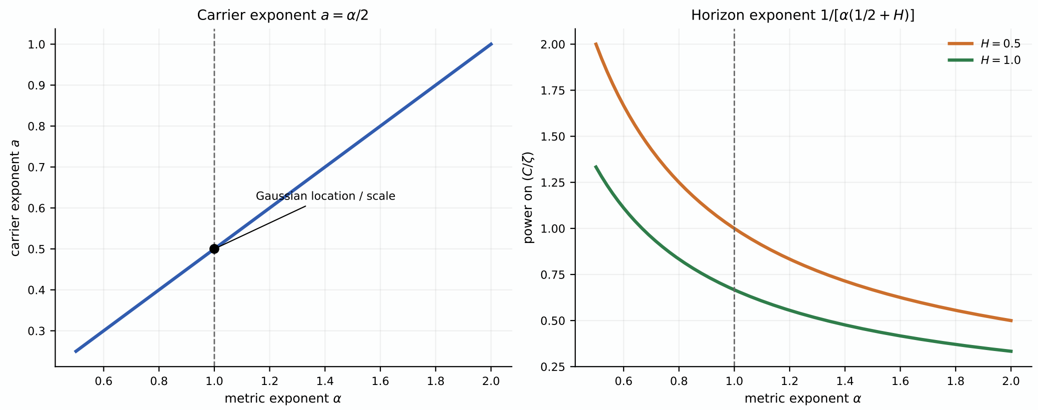

Figure 4: Regular-family inheritance beyond the canonical case. Left: if the metric geometry scales as $d(P_\theta, P_{\theta'}) \asymp \|\theta - \theta'\|^\alpha$, then the natural parametric metric carrier is $a = \alpha/2$. Right: the resulting horizon exponent becomes $1/[\alpha(1/2 + H)]$, so the canonical Gaussian-location/Gaussian-scale regime appears as the distinguished slice $\alpha = 1$ rather than as a universal rule.

Together, these settings show how the abstract law extends beyond the one-dimensional minimum kernel while keeping the formal claims within their proper scope.

Corollary 2.16 (EWMA analogue)**.** *For exponential weights $w_j = \alpha(1-\alpha)^j$, the effective lag is $\tau_\alpha = (1-\alpha)/\alpha$ and the effective sample size scales as $n_{\text{eff}} \asymp 1/\alpha$. Balancing the same carrier and staleness terms therefore yields the same exponent in the effective-memory scale. In particular, on the canonical regime $a = 1/2$, $H = 1$, the EWMA analogue of the cube-root horizon is recovered up to constants.*

The section can now be read in the order in which the theory itself develops. First comes the abstract horizon law, then its canonical closure, then its beyond-kernel extensions, and finally its prescriptive content across carrier regimes. Once a carrier regime and a roughness level are specified, the law does not merely certify that useful memory is finite; it prescribes how aggressively memory should shrink as drift intensifies and how much optimized error remains unavoidable on that regime.

3 Empirical Evidence

The empirical section makes the theoretical picture visible in finite samples. Its role is to show the main signatures of the variance–staleness law and to calibrate the operational frontier beyond the canonical proof regime.

All experiments in this section are synthetic and fully specified by the code in the repository. The Gaussian U-curve uses 20 Monte Carlo seeds; the roughness-family sweep uses 12 seeds across the stated Hölder envelopes; the fixed-ε operational frontier is calibrated from log–log carrier fits over sample sizes $(24, 48, 96, 160)$ on the ambient/intrinsic pairs $(8, 1)$, $(8, 2)$, $(12, 1)$, and $(12, 2)$ using 24 seeds; and the detector-silent- staleness experiment uses a piecewise-ramped drift stream with 12 seeds, a fixed operating window, and both ADWIN and Page–Hinkley detectors. In the operational map, a cell is marked useful when both fitted carrier exponents stay

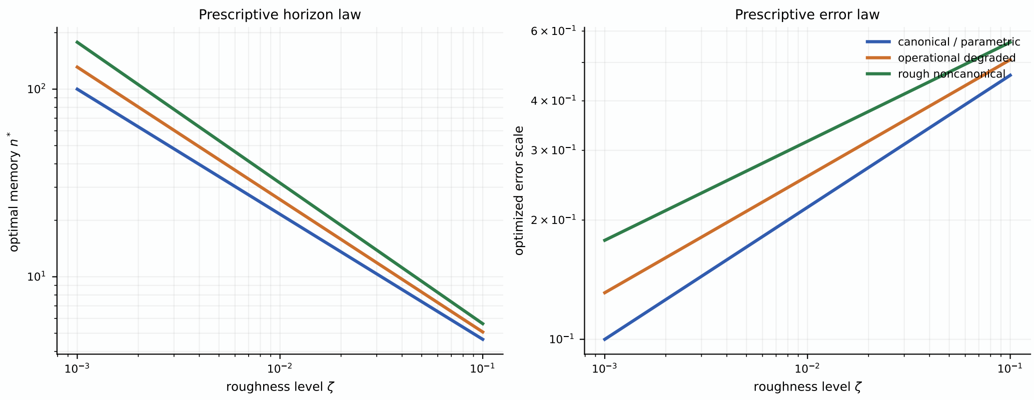

Figure 5: Prescriptive content of the horizon law. The left panel shows the optimal memory scale n^* and the right panel the optimized error scale for the $H = 1$ slice of the carrier-roughness law, across canonical, degraded operational, and rough non-canonical carrier regimes. Rougher drift and weaker carriers both push the optimal policy toward shorter memory and higher residual error.

above the usefulness threshold and their gap remains below the calibration tolerance. Reported curves are seed-averaged means, while the frontier map summarizes log–log slope fits across the listed sample sizes. The goal is not benchmark breadth but calibration of the structural signatures singled out by the theory.

Four signatures matter. First, horizon sweeps should exhibit a finite optimum rather than a boundary solution. Second, that optimum should move with temporal roughness. Third, the beyond-kernel operational regime should exhibit a visible frontier rather than a vague extension class. Fourth, temporal validity can fail before changepoint evidence becomes statistically visible.

3.1 A Visible Finite-Memory Optimum

The Gaussian horizon sweep isolates the variance–staleness trade-off directly. Small windows are variance dominated, large windows are stale, and the minimum lies at an interior horizon rather than at either boundary.

This is the most basic finite-memory signature. Under stationarity, more memory would continue to help. Under drift, the same horizon that reduces variance also ages the evidence.

3.2 Roughness-Dependent Horizon Scaling

The next experiment varies the roughness index directly. For each Hölder-type regime, the surrogate error obeys the same structural decomposition as the theory,

$$\mathcal{E}(n) \approx \sigma n^{-1/2} + \zeta n^{H},$$

and the experiment records the empirically optimal horizon across a sweep of ζ values.

This separation is structural. If the cube-root law were the only story, the right panel would collapse to a single slope. It does not. The empirical optima separate by roughness class, matching the path-dependent horizon logic of corollary 2.2.

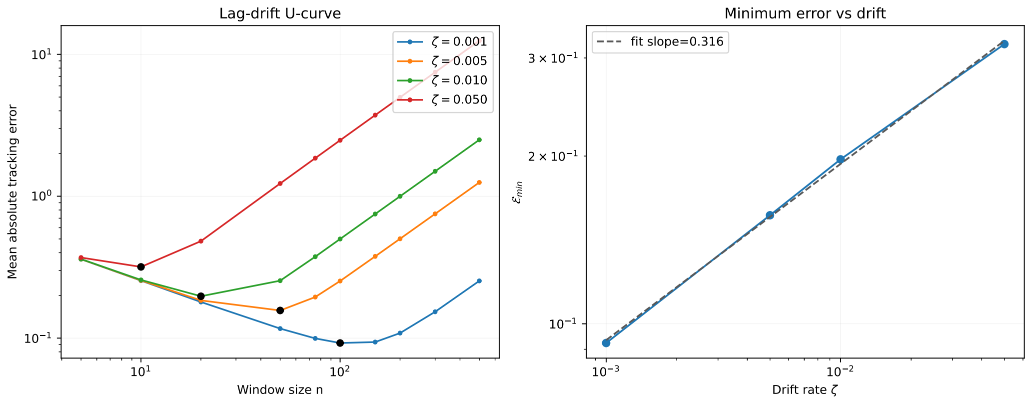

Figure 6: Structural U-curve for finite-memory tracking. Tracking error first falls with horizon and then rises once staleness overtakes the remaining variance gain. The optimum remains finite across drift strengths.

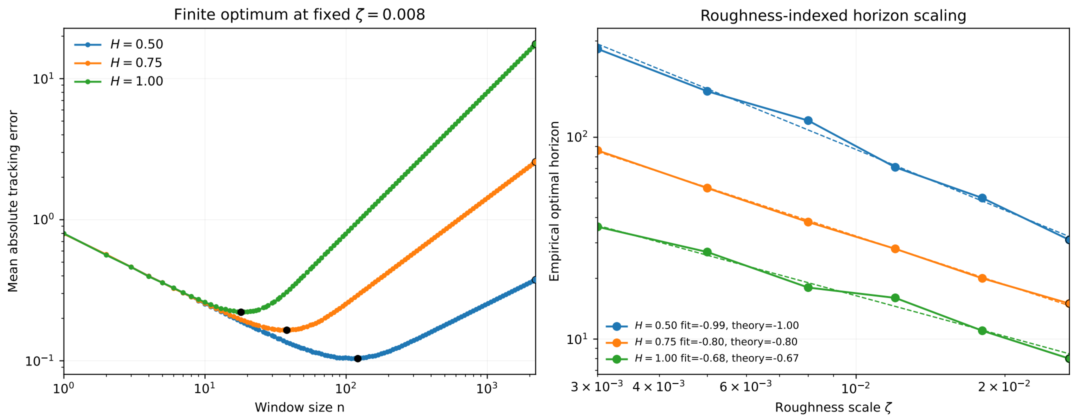

Figure 7: Roughness-dependent horizon scaling. Each path class still exhibits a finite optimum, but the location of that optimum changes with H. The worst-case Lipschitz cube-root law appears as the $H = 1$ member of a broader carrier-roughness useful-memory horizon family.

3.3 Operational Frontier Beyond the Minimum Kernel

The empirical beyond-kernel question that most directly matters for the paper is whether a practical geometry exhibits a stable carrier frontier rather than a fragile one-off phenomenon. The fixed-ε Sinkhorn experiments answer that question on the calibrated embedded fixed-span model.

The figure below concerns fixed-ε Sinkhorn geometry on embedded low-intrinsic-dimensional support. It shows that the corresponding carrier remains stable across the tested regularization values and that the triangular slope stays close to the i.i.d. benchmark rather than exhibiting a separate exponent. Combined with the structural ingredients isolated in proposition 2.12, this yields an explicit certified operational region. In the calibrated fixed-span model with $\alpha = 2$ and span 0.25, the maximal certified bands are $\varepsilon_{\max} = 0.5$ for $(d, k) = (8, 1)$, $(8, 2)$, and $(12, 2)$, and $\varepsilon_{\max} = 0.2$ for $(12, 1)$ (corollary 2.13 and fig. 3). That is the level of granularity the main story needs: a stable core, a mixed boundary case, and an operational region with visible shape rather than a flat yes-or-no claim.

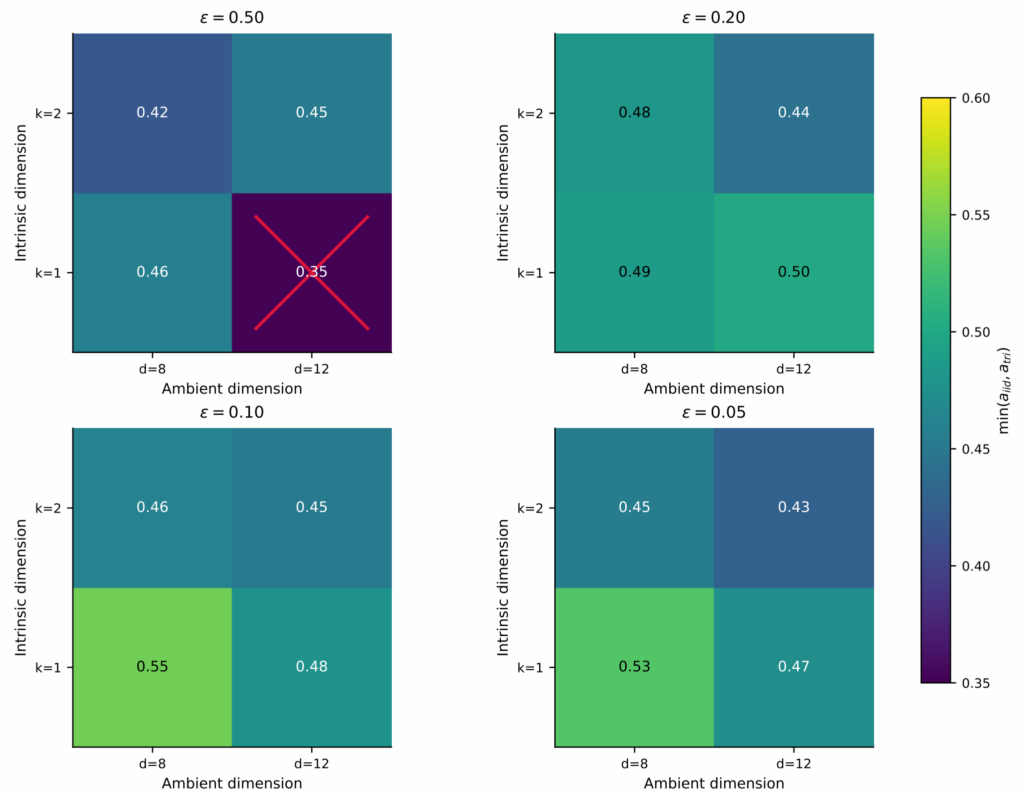

Figure 8: Empirical shape of the fixed-ε operational frontier. Each panel reports the minimum of the i.i.d. and triangular carrier exponents over the ambient/intrinsic pairs tested at one regularization level. Crossed cells mark settings that fail the usefulness criterion. The resulting picture is a frontier rather than a blanket statement: $(8, 1)$, $(8, 2)$, and $(12, 2)$ remain stable throughout the tested band, while $(12, 1)$ is the mixed boundary case.

This map adds the shape information behind the certified-frontier statement. The fixed-ε operational region is not just nonempty; it has a visible frontier, with a stable core and a single mixed boundary case on the tested grid.

3.4 Detector-Silent Staleness

Temporal validity and changepoint evidence are different clocks. To make that separation explicit, consider a stream whose local drift rate ramps upward over time. The useful-memory horizon n_t^* contracts as the ramp steepens, while a long operating horizon remains fixed. Define

$$\Delta_{\mathrm{inv}} := t_{\mathrm{detect}} - t_{\mathrm{valid}},$$

where t_{valid} is the first time the operating horizon persistently exceeds the useful-memory scale, and t_{detect} is the first detector alarm that follows.

Across the reported runs, the measured gap remains positive. This is the empirical form of detector-silent staleness: retained evidence can already be invalid for the present while detector statistics remain below alarm threshold.

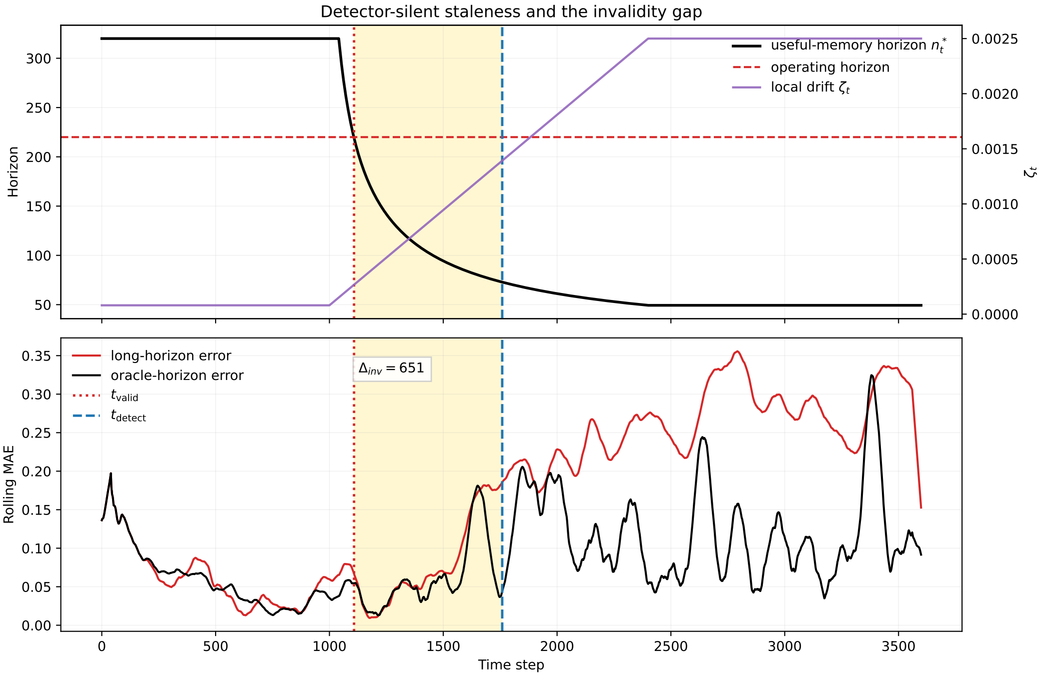

Figure 9: Detector-silent staleness. As drift intensifies, the useful-memory horizon contracts until the operating horizon becomes stale at t_{valid}; the detector responds only later at t_{detect}, leaving a strictly positive invalidity gap. The lower panel shows excess tracking error accumulating before the alarm appears.

The positive gap is not a knife-edge feature of one tuning choice. Across the tested ADWIN thresholds, mean detection time remains strictly later than mean validity-expiry time, and the same qualitative gap persists under a different detector family as well.

3.5 What the Evidence Shows

Taken together, the experiments support four claims that match the theory. Finite-memory tracking exhibits a structural U-curve rather than an ever-growing memory benefit. The useful-memory scale is roughness dependent, with the cube-root law appearing as the $H = 1$ member of a broader family. Carrier stability remains visible beyond the minimum kernel in the fixed-ε operational regime, and that evidence isolates a concrete certified region rather than a vague extension class. Temporal validity can also fail before changepoint evidence appears, producing a measurable invalidity gap.

4 Related Work

Nonstationary optimization and dynamic regret. Dynamic-regret and variation-budget results show that under drift, window size or forgetting scale must depend on how quickly the target moves [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023, Wang et al., 2024]. Those works are close in spirit, but their primary object is regret under a moving comparator. Here the primary object is different: the temporal validity of retained evidence in a distribution-tracking problem. The horizon is not a tuning parameter attached to regret analysis; it is the statistical object being characterized.

Adaptive windowing and drift detection. Adaptive-window and concept-drift methods ask when accumulated evidence is sufficient to react [Bifet and Gavaldà, 2007, Gama et al., 2004, 2014]. Hinder et al. [2023] emphasize that the relevant window scale is itself part of the problem. The present question is adjacent but distinct. Temporal validity is not changepoint evidence: memory can stop representing the present before the available data support a detector alarm. In that sense, the paper is about the validity horizon of retained evidence rather than the stopping time of a reaction rule. The invalidity gap makes that distinction visible in finite samples.

Adaptive filtering and state-space tracking. Classical adaptive filtering and state-space tracking also balance smoothing against responsiveness by tuning forgetting factors, gain schedules, or process noise models [Kalman, 1960, Jazwinski, 1970]. Those analyses usually work under stochastic state-space assumptions and ask for Bayes-optimal tracking of a latent process. The present paper instead studies deterministic Wasserstein drift envelopes and characterizes a worst-case temporal-validity horizon.

Locally stationary smoothing. Locally stationary time-series analysis likewise studies how window or bandwidth choice must adapt to time variation [Dahlhaus, 1997]. That literature is close in its bias–variance logic, but the moving object there is typically a local spectrum or regression function. Here the moving object is a path of distributions, and the age penalty is measured directly in transport geometry.

Distribution tracking under transport geometry. Wasserstein distance and Sinkhorn-type estimators are natural when drift lives on paths of distributions rather than on scalar losses [Fournier and Guillin, 2015, Genevay et al., 2019, Mena and Weed, 2019, Stromme, 2023, Groppe and Hundrieser, 2024, Niles-Weed and Bach, 2019, Goldfeld et al., 2024, Akyildiz et al., 2024]. They supply the geometric language in which staleness becomes a transport error between past and present. In this setting, the trade-off is not simply between adaptation and variance, but between finite-sample accuracy and temporal misalignment measured in distributional geometry.

Entropic and smoothed transport as operational carriers. Regularization changes the statistical geometry as well as the computational one. Entropic and other smoothed transport quantities introduce explicit bias–variance–regularization trade-offs and can recover more stable statistical behavior than raw high-dimensional Wasserstein distances [Genevay et al., 2019, Goldfeld et al., 2024, Stromme, 2023, Groppe and Hundrieser, 2024, Akyildiz et al., 2024]. That perspective is exactly why the paper treats fixed-ε Sinkhorn as an operational carrier rather than as a surrogate for the raw W_2 theorem. The point is not that regularization preserves the canonical geometry unchanged, but that it can induce a usable carrier regime on which a finite-memory horizon law still makes sense.

Wasserstein nonstationarity and robust optimization. Jiang, Li, and Zhang study online stochastic optimization under a Wasserstein nonstationarity measure and obtain distribution-aware regret guarantees [Jiang et al., 2020]. Keehan, Anderson, and Wiesemann analyze related drift-aware weighting questions in Wasserstein distributionally robust optimization [Keehan et al., 2025]. Amini and Vinas give a recent moment-method route to triangular-array concentration in W_1 [Amini and Vinas, 2026]. Those results are adjacent to the carrier-regime issue studied here, but they do not by themselves close the present W_2 window-mixture concentration problem. The present focus is narrower and more structural: for finite-memory tracking, how long can retained evidence continue to improve estimation of the current distribution before age becomes a dominant error source?

Freshness and Age of Information. The Age-of-Information literature treats age as a resource constraint and asks how update policies balance communication cost against stale information [Kaul et al., 2012, Yates et al., 2021]. The present paper gives a distributional analogue. Age is costly because it induces geometric error under drift, and temporal roughness determines how quickly that cost accumulates.

Across these neighboring literatures, the closest theme is that nonstationarity makes retention scale matter. The distinguishing claim here is more specific: finite memory obeys a carrier-roughness useful-memory horizon law, the worst-case cube-root law is the $a = 1/2$, $H = 1$ member of that family, and temporal validity can fail before change becomes statistically detectable. The resulting viewpoint is complementary to regret and detection analyses: it characterizes when memory itself ceases to be a statistically valid proxy for the present.

5 Limitations and Scope

The central results of the paper are the carrier-roughness useful-memory horizon law, the minimum-kernel carrier on the canonical $a = 1/2$ regime, and the structural lower bound on that same regime. The refined Gaussian witness results sharpen the constants inside that regime, but broader extensions are stated more cautiously.

Carrier scope. The theory is modular in the carrier exponent a, but the main rigorous regime is the root-n carrier. The minimum kernel establishes that regime in one dimension under bounded support and fixed span. Outside that setting, the same balancing logic continues to hold, but the carrier must come from a different theorem. In particular, the paper does not claim a general raw-W_2 triangular-array carrier theorem in arbitrary dimension.

Extended and operational regimes. The extended setting and the fixed-ε Sinkhorn setting are supported here differently. The extended setting remains primarily finite-sample evidence. For fixed-ε Sinkhorn, the paper now isolates a certified frontier: exact support-complexity inheritance and the dual threshold $\alpha > k/2$ are structural on the embedded fixed-span model, while the triangular-vs.-benchmark carrier comparison is still supplied by calibration on the tested ε-bands. A theorem of the same strength as the minimum-kernel result is therefore still not established here, but the remaining gap is narrower and explicit.

Distributional extension class. The paper does not support a blanket extension to every deterministic W_2-Hölder path class. Continuous regular families and operational geometries can support meaningful carrier exponents, but support-changing and singular families behave differently. The right noncanonical target is therefore a theorem for regular dominated families or operational geometries with comparable local testing structure, not a universal statement over arbitrary transport-Hölder paths.

Lower-bound scope. The lower bound is subclass based. The Gaussian witness is enough to show that on the canonical carrier regime $a = 1/2$, the useful-memory scale is structural rather than an artifact of a particular estimator. What remains open is class-tightness and constant-sharpness for full deterministic Hölder path classes, especially away from the canonical regime and beyond the witness geometry developed here. The witness frontier is already sharper than the original ramp-plus-Pinsker sketch: the exact Gaussian testing bound improves the canonical-regime constant, and for $H < 1$ the endpoint-minimal Hölder witness is stronger than the ramp witness. But those are still witness-level refinements. The paper now includes a full minimax rate statement for the deterministic Hölder class in the Gaussian location benchmark, but constant-sharp and fully distributional class-tight theorems remain open. The refined proof-gap consequence remains at that witness-constant level: it clarifies the residual geometry inside the exact Gaussian witness program, not the last word on the full distributional Hölder class.

Weights and memory shape. The main text emphasizes uniform windows, with an EWMA corollary showing that the same exponent logic survives after effective-lag reparameterization. The paper does not investigate whether more general non-uniform weights can improve constants or alter exponents across broader path classes.

Online roughness estimation. The law is stated in terms of the roughness parameters H and ζ, but the paper does not propose a fully online procedure for estimating them or for quantifying the sensitivity of the chosen horizon to misspecification. Turning the prescriptive law into a robust adaptive policy remains future work.

Empirical scope. The experiments are synthetic and structurally motivated. Their role is to make the theory visible and to calibrate broader carrier settings, not to demonstrate competitive performance on a benchmark suite. Real-data validation and direct comparison with adaptive horizon-selection rules remain outside the present scope. The paper also does not include dedicated changepoint-detection baselines such as IDD or sRE, nor a benchmark study across alternative window-selection heuristics.

Detectability versus validity. Detector-silent staleness is an empirical consequence of the structural theory, not the primary theorem object. The paper shows that temporal validity can fail before detector alarms appear, but it does not prove a universal detection-delay theorem across detector classes.

Open problems. The next technical questions are clear. The first is to establish carrier inheritance beyond the minimum kernel under natural low-dimensional conditions. The second is to understand when practical geometries, such as fixed- ε Sinkhorn, admit full triangular-array inheritance theorems that remove the present empirical calibration step in genuinely mid- or high-dimensional settings. The third is to build the broader family theory: determine which regular dominated distribution families support which carrier exponents a, and then extend the lower bound beyond the canonical $a = 1/2$ regime accordingly. After that comes the question of how far the new Gaussian-witness refinements can be pushed toward class-tight lower theory. After that come non-uniform weighting, real-data validation, and stronger stochastic-process formulations.

6 Conclusion

Under drift, memory is never purely beneficial. It reduces finite-sample noise, but it also ages. The resulting trade-off makes useful memory a finite-horizon object rather than an unlimited statistical resource.

The paper formalizes that idea through a carrier-roughness useful-memory horizon law. The central theorem is the abstract decomposition

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, P_t\Big) \leq C_K n^{-a} + C_S \zeta n^H,$$

whose optimized scale is

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

The cube-root horizon is therefore one member of a broader law: the special case $a = 1/2$, $H = 1$ corresponding to a root-n carrier under worst-case linear staleness.

The analysis anchors this law in a rigorous minimum-kernel result and a structural lower bound on the same canonical carrier regime. Together they show that, in the bounded-support fixed-span one-dimensional setting, the horizon is neither an artifact of a particular estimator nor a purely heuristic balance. The refined Gaussian witness analysis sharpens that conclusion: exact testing gives a clean scalar frontier, and the endpoint-minimal Hölder witness improves the ramp constant for $H < 1$. In the Gaussian location benchmark, the same canonical geometry also yields a full minimax rate statement on the deterministic Hölder class. Beyond that proof kernel, the paper isolates two main directions. On the operational side, fixed-ε Sinkhorn is now organized by a certified frontier: exact support-complexity inheritance, the dual threshold $\alpha > k/2$, and calibrated band stability together identify the region on which the operational carrier inherits the horizon law of its i.i.d. benchmark. On the distributional side, the right extension target is no longer a false universal root-n theorem, but a family-specific horizon principle in which regular families support different carrier exponents and hence different members of the horizon family.

The empirical section mirrors that sequence. It shows a structural U-curve, roughness-dependent horizon scaling, a visible operational frontier, and a strictly positive invalidity gap between temporal validity and changepoint evidence. The resulting message is both theoretical and operational: useful memory has a path-dependent horizon, and retained evidence can become invalid before change becomes statistically detectable.

The resulting shift in viewpoint is that memory under drift is not treated primarily as a regret-tuning parameter or an alarm-design parameter, but as a temporal-validity object in distributional geometry.

The main open questions concern carrier inheritance beyond the minimum kernel, theorem-level operational-geometry results, and broader lower-bound extensions beyond the canonical $a = 1/2$ regime. These are natural extensions of the present theory, not changes to its central object.

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